# SC1001: Introduction to Computational Thinking

## Tutorial 1: Flowcharts, Pseudocode, Data Types, and Variables

Jordan Boyd-Graber

Nanyang Technological University

# Me

- Professor in CCDS
- Formerly at the University of Maryland and the University of Colorado
- Research: natural language processing, question answering, topic models, and human–AI interaction
- Born in Colorado; grew up in Iowa
- High school in Arkansas
- Undergraduate study in California
- Graduate school in New Jersey
- Worked on an electronic dictionary in Berlin and Google Books in New York
- You can call me Jordan, JBG, or Boyd-Graber

# What We Are Practising Today

- Trace and repair an algorithm
- If statements
- FizzBuzz
- Distinguish Python expressions, assignments, and errors
- Design an algorithm before writing Python
- Use a simple heuristic when an exact solution is unnecessary

# Discussion 1: What Is Wrong?

The algorithm should read 10 scores and compute their average.

```
Initialize student_counter to zero
While student_counter is less than or equal to ten
    Input the next score
    Add the score into the total
EndWhile
Set class_average to total divided by ten
```

Find every variable or loop-control error before proposing a repair.

# Discussion 1: What Is Wrong?

The algorithm should read 10 scores and compute their average.

```
Initialize student_counter to zero
While student_counter is less than or equal to ten
    Input the next score
    Add the score into the total
EndWhile
Set class_average to total divided by ten
```

What was this originally?

# Discussion 1: What Is Wrong?

The algorithm should read 10 scores and compute their average.

```
Initialize student_counter to zero
While student_counter is less than or equal to ten
    Input the next score
    Add the score into the total
EndWhile
Set class_average to total divided by ten
```

Will this loop end?

# Discussion 1: A Corrected Algorithm

```
Set total to 0
Set student_counter to 1
While student_counter <= 10
    Input score
    Set total to total + score
    Set student_counter to student_counter + 1
EndWhile
Set class_average to total / 10
```

**Three issues**

Initialise before use

# Discussion 1: A Corrected Algorithm

```
Set total to 0
Set student_counter to 1
While student_counter <= 10
    Input score
    Set total to total + score
    Set student_counter to student_counter + 1
EndWhile
Set class_average to total / 10
```

**Three issues**

Process exactly ten scores

# Discussion 1: A Corrected Algorithm

```
Set total to 0
Set student_counter to 1
While student_counter <= 10
    Input score
    Set total to total + score
    Set student_counter to student_counter + 1
EndWhile
Set class_average to total / 10
```

**Three issues**

Update the loop-control variable

# Discussion 1: An Alternate Version

```
Set total to 0
Set student_counter to 0
While student_counter < 10
    Input score
    Set total to total + score
    Set student_counter to student_counter + 1
EndWhile
Set class_average to total / student_counter
```

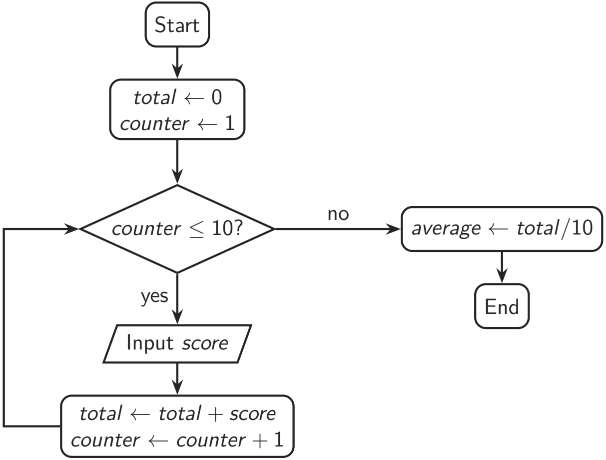
# Discussion 1: An Alternate Version

```
Set total to 0
Set student_counter to 0
While student_counter < 10
    Input score
    Set total to total + score
    Set student_counter to student_counter + 1
EndWhile
Set class_average to total / student_counter
```

**Can you tell my first language was C?**

I don't like magic numbers in the code. You might change how many students you count, or let a user terminate early, and this would break the assumption that you have 10 students.
Another reason to do this is that the `range` command in Python will give you numbers less than the argument.

# Discussion 1: The Control Flow



# Discussion 2: Fill in the Decisions

```
Set passes to 0; set failures to 0
Set student_counter to 1
While student_counter <= 10
    Input score

    --------------------------
        ----------------------
    Else

        ----------------------
    EndIf
    Set student_counter to student_counter + 1
EndWhile

------------------------------
```

Assume the pass mark is available as `pass_mark`.

# Discussion 2: One Solution

```
If score >= pass_mark
    Set passes to passes + 1
Else
    Set failures to failures + 1
EndIf
...
Print passes
Print failures
```

**Check**

After every iteration, $passes + failures = student\_counter - 1$.

# Discussion 3: FizzBuzz

For each integer from 1 through 20:

- print `FizzBuzz` if it is divisible by both 3 and 5;
- otherwise print `Fizz` if divisible by 3;
- otherwise print `Buzz` if divisible by 5;
- otherwise print the number.

# Discussion 3: FizzBuzz

For each integer from 1 through 20:

- print `FizzBuzz` if it is divisible by both 3 and 5;
- otherwise print `Fizz` if divisible by 3;
- otherwise print `Buzz` if divisible by 5;
- otherwise print the number.

Hint: Order matters!

# Discussion 3: FizzBuzz

For each integer from 1 through 20:

- otherwise print `Fizz` if divisible by 3;
- otherwise print `Buzz` if divisible by 5;
- otherwise print the number.

When given as an interview problem, this step is often left out, you're supposed to ask.

# Discussion 3: Pseudocode

```
For num from 1 to 20
    If num MOD 15 = 0
        Print "FizzBuzz"
    Else if num MOD 3 = 0
        Print "Fizz"
    Else if num MOD 5 = 0
        Print "Buzz"
    Else
        Print num
    EndIf
EndFor
```

# Discussion 3: Full Python Solution

```python
for num in range(1, 21):
    if num % 15 == 0:
        print("FizzBuzz")
    elif num % 3 == 0:
        print("Fizz")
    elif num % 5 == 0:
        print("Buzz")
    else:
        print(num)
```

# Discussion 3: Modulo in Math Notation

If $a \bmod n = r$, then dividing $a$ by $n$ leaves remainder $r$.

$$15 \bmod 3 = 0 \qquad 15 \bmod 5 = 0$$

$$14 \bmod 3 = 2 \qquad 14 \bmod 5 = 4$$

## Number-theory notation

$$n \equiv 0 \pmod{3} \quad \text{means} \quad 3 \mid n$$

$$n \equiv 0 \pmod{5} \quad \text{means} \quad 5 \mid n$$

$$n \equiv 0 \pmod{15} \quad \text{means} \quad 15 \mid n$$

So FizzBuzz starts by testing whether $n \equiv 0 \pmod{15}$ before the separate tests modulo 3 and 5.

# Can we do it with two if statements?

```
for num in range(1, 20):
     if num % 3 == 0:
         print("Fizz", end="")
     if num % 5 == 0:
         print("Buzz", end="")
     if (not num % 3 == 0) and (not num % 5 ==0):
         print(num, end="")
     print("!")
```

Tricky because Python automatically puts newline after `print` command.

# Discussion 4: Predict Before Running (1–9)

Assume the lines are executed in this order.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

c = 10: valid assignment; now c stores 10.

# Discussion 4: Predict Before Running (1–9)

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b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

7 = a: syntax error because the left side is not assignable.

# Discussion 4: Predict Before Running (1–9)

Assume the lines are executed in this order.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`a = d`: name error because `d` has not been defined.

# Discussion 4: Predict Before Running (1–9)

Assume the lines are executed in this order.

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c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
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import = 1003
b = math.pi * c
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```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`a = c + 1`: valid assignment; a becomes 11 if line 1 already ran.

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b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`a + c = c`: syntax error because an expression cannot be assigned to.

# Discussion 4: Predict Before Running (1–9)

Assume the lines are executed in this order.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`3 + a`: valid expression (14), but a script prints nothing unless asked.

# Discussion 4: Predict Before Running (1–9)

Assume the lines are executed in this order.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

7up = 10: syntax error because identifiers cannot start with a digit.

# Discussion 4: Predict Before Running (1–9)

Assume the lines are executed in this order.

```
c = 10
7 = a
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a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`import = 1003`: syntax error because `import` is a keyword.

# Discussion 4: Predict Before Running (1–9)

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c = 10
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a + c = c
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b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`b = math.pi * c`: name error unless `math` was imported first, otherwise 31.415...

# Discussion 4: Predict Before Running (10–18)

Assume the lines are executed in this order.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`int = 500`: legal, but it shadows the built-in `int`.

# Discussion 4: Predict Before Running (10–18)

Assume the lines are executed in this order.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`a ** 3`: valid expression if a exists (1331); no printed output in a script.

# Discussion 4: Predict Before Running (10–18)

Assume the lines are executed in this order.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`a, b, c = c, 1, a`: valid simultaneous assignment if the names already exist:
$a = 10$; $b = 1$; $c = 11$

# Discussion 4: Predict Before Running (10–18)

Assume the lines are executed in this order.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`b, c, a = a, b`: value error because three variables expect only two values.

# Discussion 4: Predict Before Running (10–18)

Assume the lines are executed in this order.

```
c = 10
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a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
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a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`c = b = a = 7`: valid chained assignment: $c = 7, b = 7, a = 7$

# Discussion 4: Predict Before Running (10–18)

Assume the lines are executed in this order.

```
c = 10
7 = a
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a + c = c
3 + a
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b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`print(A)`: name error because Python is case-sensitive.

# Discussion 4: Predict Before Running (10–18)

Assume the lines are executed in this order.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
3 + a
7up = 10
import = 1003
b = math.pi * c
```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`print("b*b + a*a = c*c")`: prints the literal text, not a calculation.

# Discussion 4: Predict Before Running (10–18)

Assume the lines are executed in this order.

```
c = 10
7 = a
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```

```
int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`print('A')`: prints the character `A`.

# Discussion 4: Predict Before Running (10–18)

Assume the lines are executed in this order.

```
c = 10
7 = a
a = d
a = c + 1
a + c = c
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int = 500
a ** 3
a, b, c = c, 1, a
b, c, a = a, b
c = b = a = 7
print(A)
print("b*b + a*a = c*c")
print('A')
print("c" = 1)
```

`print("c" = 1)`: syntax error because assignment is not a function argument.

# Discussion 5: Class Percentages

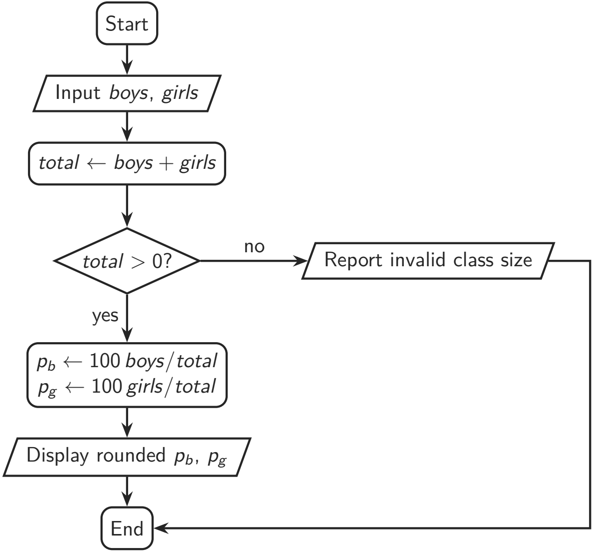
Ask for the numbers of boys and girls, then display each percentage.

**Sample interaction**

```
Enter the number of boys:  65
Enter the number of girls:  77
Boys:  46%    Girls:  54%
```

Design the algorithm and flowchart first.

# Discussion 5: Flowchart



# Discussion 5: Python

```
boys = int(input("Enter the number of boys: "))
girls = int(input("Enter the number of girls: "))
total = boys + girls

if total <= 0:
    print("The class must contain at least one student.")
else:
    percent_boys = boys / total
    percent_girls = girls / total
    print("Boys:", format(percent_boys, ".0%"))
    print("Girls:", format(percent_girls, ".0%"))
```

# Discussion 6: Route Planning

Start at the hotel, visit every attraction once, and return to the hotel.

- We want a short route, but do not need the absolute optimum.
- Assume a distance matrix already exists.
- Propose a logical, easy-to-follow heuristic in pseudocode.

# Discussion 6: Route Planning

Start at the hotel, visit every attraction once, and return to the hotel.

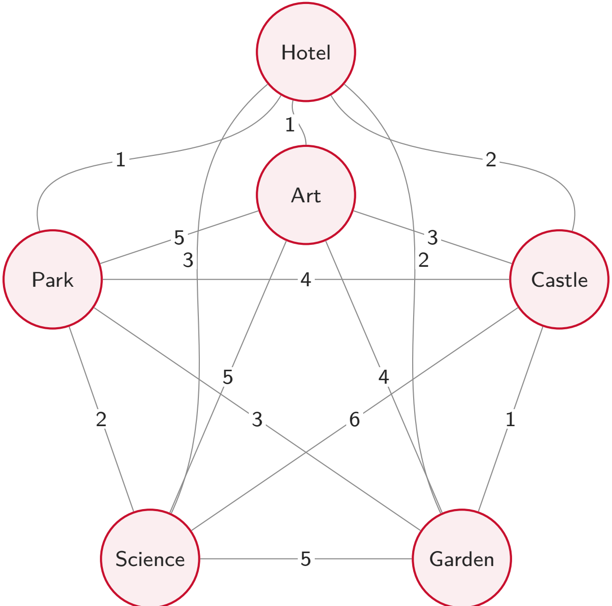
- We want a short route, but do not need the absolute optimum.
- Assume a distance matrix already exists.
- Propose a logical, easy-to-follow heuristic in pseudocode.

Propose a solution to the above problem and describe it using pseudocode. You do NOT need to find the absolute best route, ignore data structure details; assume that the distance matrix has already been created.

# Discussion 6: Distances Between Attractions

| | Hotel | Art | Castle | Garden | Science | Park |
|---|---|---|---|---|---|---|
| Hotel | 0 | 1 | 2 | 2 | 3 | 1 |
| Museum of Art | 1 | 0 | 3 | 4 | 5 | 5 |
| Historical Castle | 2 | 3 | 0 | 1 | 6 | 4 |
| Botanical Garden | 2 | 4 | 1 | 0 | 5 | 3 |
| Science Center | 3 | 5 | 6 | 5 | 0 | 2 |
| City Park | 1 | 5 | 4 | 3 | 2 | 0 |

# Discussion 6: The Distance Graph



# Discussion 6: Nearest-Neighbour Heuristic

```
Set current_location to Hotel
Mark every attraction unvisited; set route to [Hotel]
While an unvisited attraction remains
    Set nearest_distance to infinity
    For each attraction in unvisited
        Set distance to distance_matrix[current_location][attraction]
        If distance < nearest_distance
            Set nearest_distance to distance
            Set nearest_attraction to attraction
        EndIf
    EndFor
    Visit and mark nearest_attraction; append it to route
    Set current_location to nearest_attraction
EndWhile
Append Hotel to route; display route
```

Repeat until we've visited everything

# Discussion 6: Nearest-Neighbour Heuristic

```
Set current_location to Hotel
Mark every attraction unvisited; set route to [Hotel]
While an unvisited attraction remains
    Set nearest_distance to infinity
    For each attraction in unvisited
        Set distance to distance_matrix[current_location][attraction]
        If distance < nearest_distance
            Set nearest_distance to distance
            Set nearest_attraction to attraction
        EndIf
    EndFor
    Visit and mark nearest_attraction; append it to route
    Set current_location to nearest_attraction
EndWhile
Append Hotel to route; display route
```

Check every currently unvisited attraction.

# Discussion 6: Nearest-Neighbour Heuristic

```
Set current_location to Hotel
Mark every attraction unvisited; set route to [Hotel]
While an unvisited attraction remains
    Set nearest_distance to infinity
    For each attraction in unvisited
        Set distance to distance_matrix[current_location][attraction]
        If distance < nearest_distance
            Set nearest_distance to distance
            Set nearest_attraction to attraction
        EndIf
    EndFor
    Visit and mark nearest_attraction; append it to route
    Set current_location to nearest_attraction
EndWhile
Append Hotel to route; display route
```

Update the best candidate only when we find a shorter edge.

# Discussion 6: Nearest-Neighbour Heuristic

```
Set current_location to Hotel
Mark every attraction unvisited; set route to [Hotel]
While an unvisited attraction remains
    Set nearest_distance to infinity
    For each attraction in unvisited
        Set distance to distance_matrix[current_location][attraction]
        If distance < nearest_distance
            Set nearest_distance to distance
            Set nearest_attraction to attraction
        EndIf
    EndFor
    Visit and mark nearest_attraction; append it to route
    Set current_location to nearest_attraction
EndWhile
Append Hotel to route; display route
```

Save that stop.

# Discussion 6: Nearest-Neighbour Heuristic

```
Set current_location to Hotel
Mark every attraction unvisited; set route to [Hotel]
While an unvisited attraction remains
    Set nearest_distance to infinity
    For each attraction in unvisited
        Set distance to distance_matrix[current_location][attraction]
        If distance < nearest_distance
            Set nearest_distance to distance
            Set nearest_attraction to attraction
        EndIf
    EndFor
    Visit and mark nearest_attraction; append it to route
    Set current_location to nearest_attraction
EndWhile
Append Hotel to route; display route
```

Save choice and repeat.

# Discussion 6: Ties and Trade-offs

Hotel–Art and Hotel–Park are both 1 km.

- Use a deterministic tie-breaker: first in the list, alphabetical order, or lowest ID.
- The heuristic is fast and easy to explain.
- It can miss the globally shortest tour because it never revisits an earlier choice.

**One deterministic run**

Hotel → Art → Castle → Garden → Park → Science → Hotel
Total distance: $1 + 3 + 1 + 3 + 2 + 3 = 13$ km.

# Discussion 6: Ties and Trade-offs

Hotel–Art and Hotel–Park are both 1 km.

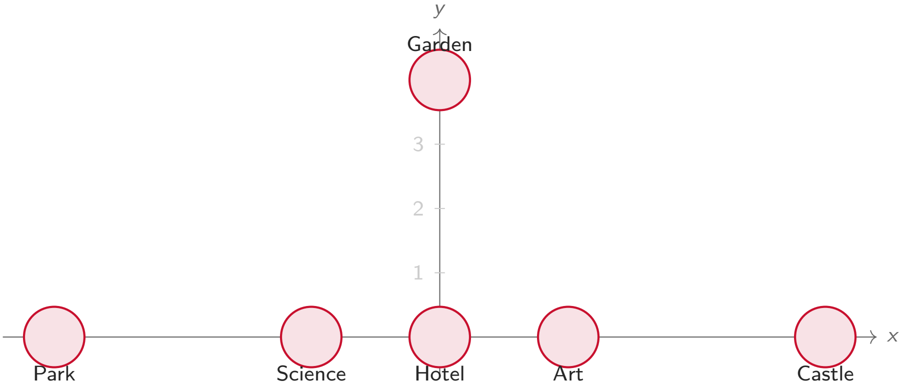
- Use a deterministic tie-breaker: first in the list, alphabetical order, or lowest ID.
- The heuristic is fast and easy to explain.
- It can miss the globally shortest tour because it never revisits an earlier choice.

**One deterministic run**

Hotel $\rightarrow$ Art $\rightarrow$ Castle $\rightarrow$ Garden $\rightarrow$ Park $\rightarrow$ Science $\rightarrow$ Hotel
Total distance: $1 + 3 + 1 + 3 + 2 + 3 = 13$ km.

Can you think of where the Greedy approach might fail?

# Discussion 6: A Counterexample in the Plane




Points: Hotel $(0,0)$, Art $(1,0)$, Castle $(3,0)$, Garden $(0,4)$, Science $(-1,0)$, Park $(-3,0)$.

# Discussion 6: Counterexample Distance Matrix

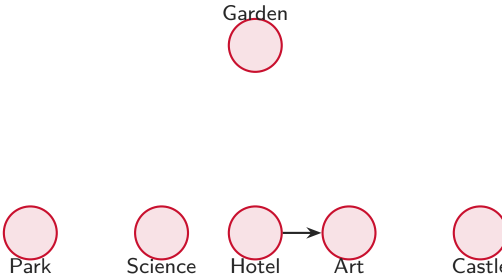
| | Hotel | Art | Castle | Garden | Science | Park |
|---|---|---|---|---|---|---|
| Hotel | 0.00 | 1.00 | 3.00 | 4.00 | 1.00 | 3.00 |
| Art | 1.00 | 0.00 | 2.00 | 4.12 | 2.00 | 4.00 |
| Castle | 3.00 | 2.00 | 0.00 | 5.00 | 4.00 | 6.00 |
| Garden | 4.00 | 4.12 | 5.00 | 0.00 | 4.12 | 5.00 |
| Science | 1.00 | 2.00 | 4.00 | 4.12 | 0.00 | 2.00 |
| Park | 3.00 | 4.00 | 6.00 | 5.00 | 2.00 | 0.00 |

These are Euclidean distances rounded to two decimal places.

# Discussion 6: Walking Through the Greedy Choice

Assume ties are broken alphabetically, so from Hotel the algorithm chooses Art instead of Science.

- Hotel $\rightarrow$ Art because both Art and Science are distance 1.00.

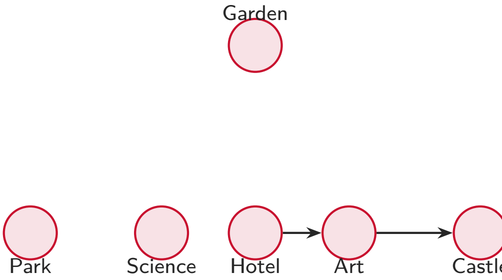



Running total grows as the path commits to each local choice.

# Discussion 6: Walking Through the Greedy Choice

Assume ties are broken alphabetically, so from Hotel the algorithm chooses Art instead of Science.

- Hotel → Art because both Art and Science are distance 1.00.
- Art → Castle because 2.00 is the smallest remaining distance.

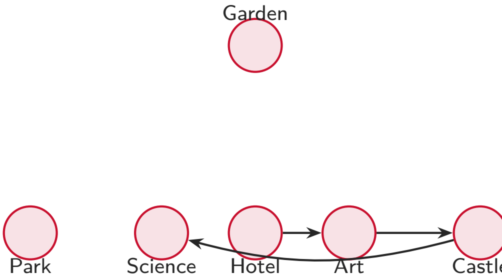



Running total grows as the path commits to each local choice.

# Discussion 6: Walking Through the Greedy Choice

Assume ties are broken alphabetically, so from Hotel the algorithm chooses Art instead of Science.

- Hotel → Art because both Art and Science are distance 1.00.
- Art → Castle because 2.00 is the smallest remaining distance.
- Castle → Science because 4.00 beats 5.00 and 6.00.

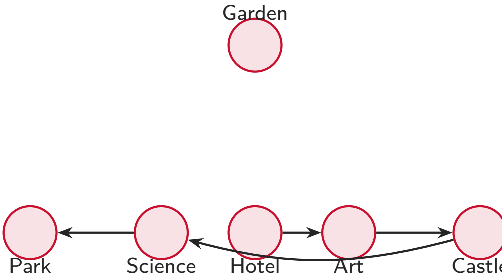



Running total grows as the path commits to each local choice.

# Discussion 6: Walking Through the Greedy Choice

Assume ties are broken alphabetically, so from Hotel the algorithm chooses Art instead of Science.

- Hotel $\rightarrow$ Art because both Art and Science are distance 1.00.
- Art $\rightarrow$ Castle because 2.00 is the smallest remaining distance.
- Castle $\rightarrow$ Science because 4.00 beats 5.00 and 6.00.
- Science $\rightarrow$ Park because 2.00 is smallest.



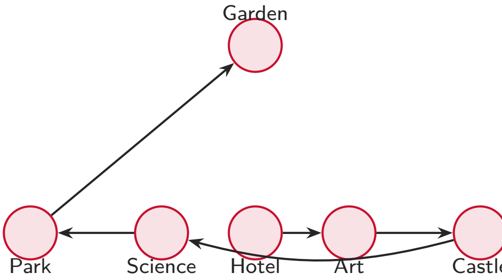

Running total grows as the path commits to each local choice.

# Discussion 6: Walking Through the Greedy Choice

Assume ties are broken alphabetically, so from Hotel the algorithm chooses Art instead of Science.

- Hotel $\rightarrow$ Art because both Art and Science are distance 1.00.
- Art $\rightarrow$ Castle because 2.00 is the smallest remaining distance.
- Castle $\rightarrow$ Science because 4.00 beats 5.00 and 6.00.
- Science $\rightarrow$ Park because 2.00 is smallest.
- Park $\rightarrow$ Garden is forced, with distance 5.00.

Running total grows as the path commits to each local choice.

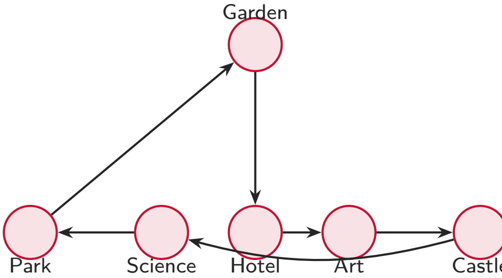


# Discussion 6: Walking Through the Greedy Choice

Assume ties are broken alphabetically, so from Hotel the algorithm chooses Art instead of Science.

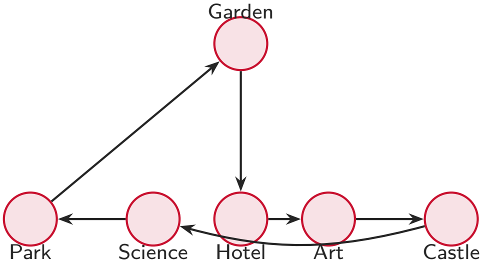
- Hotel $\rightarrow$ Art because both Art and Science are distance 1.00.
- Art $\rightarrow$ Castle because 2.00 is the smallest remaining distance.
- Castle $\rightarrow$ Science because 4.00 beats 5.00 and 6.00.
- Science $\rightarrow$ Park because 2.00 is smallest.
- Park $\rightarrow$ Garden is forced, with distance 5.00.
- Garden $\rightarrow$ Hotel closes the tour.

Total greedy length:
$1.00 + 2.00 + 4.00 + 2.00 + 5.00 + 4.00 = 18.00.$



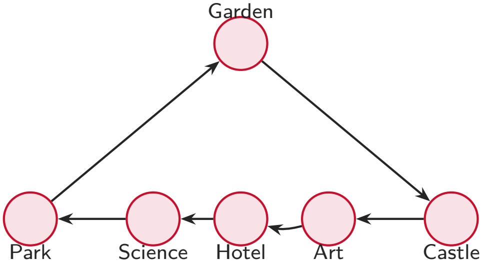
# Discussion 6: Greedy vs. Better Tour

**Greedy choice**




Hotel → Art → Castle → Science → Park → Garden → Hotel
18.00

**Shorter tour**




Hotel → Science → Park → Garden → Castle → Art → Hotel
16.00

# Discussion 6: Python Code (1/2)

```python
distances = {
    "Hotel": {"Art": 1, "Castle": 2, "Garden": 2, "Science": 3, "Park": 1},
    "Art": {"Hotel": 1, "Castle": 3, "Garden": 4, "Science": 5, "Park": 5},
    "Castle": {"Hotel": 2, "Art": 3, "Garden": 1, "Science": 6, "Park": 4},
    "Garden": {"Hotel": 2, "Art": 4, "Castle": 1, "Science": 5, "Park": 3},
    "Science": {"Hotel": 3, "Art": 5, "Castle": 6, "Garden": 5, "Park": 2},
    "Park": {"Hotel": 1, "Art": 5, "Castle": 4, "Garden": 3, "Science": 2},
}
```

# Discussion 6: Python Code (2/2)

```
route = ["Hotel"]
unvisited = {"Art", "Castle", "Garden", "Science", "Park"}
current = "Hotel"

while unvisited:
    nearest_distance = float("inf")
    nearest_attraction = None

    for place in unvisited:
        distance = distances[current][place]
        if distance < nearest_distance:
            nearest_distance = distance
            nearest_attraction = place

    route.append(nearest_attraction)
    unvisited.remove(nearest_attraction)
    current = nearest_attraction

route.append("Hotel")
print(" -> ".join(route))
```

# Takeaways

- Initialize variables
- Don't overcomplicate expressions
- Good enough isn't always optimal
- Flowcharts expose decisions and loops before code obscures them.